CS801 – Internet of Things

Unit – I

- **Main**
  - IoT Basics
  - IoT Architecture
  - IoT Network Design
- **Detail**
  - IoT definition – Characteristics – Conceptual framework
  - IoT ecosystem components
  - Physical – Logical design of IoT
  - IoT enablers – Modern IoT applications
  - M2M communication – IoT vs M2M – IoT vs WoT
  - IoT reference architecture
  - IoT network configuration – IoT LAN – IoT WAN
  - IoT Node – Gateway – Proxy
  - Basic microcontrollers and interfacing

Unit – II

- **Main**
  - Sensors
  - Sensor Errors
  - Actuators
- **Detail**
  - Sensor definition – Sensor node components – Challenges
  - Sensor features – Resolution
  - Sensor classes (Analog – Digital – Scalar – Vector)
  - Sensor errors (Bias – Drift – Hysteresis – Quantization)
  - Actuator definition
  - Actuator types (Hydraulic – Pneumatic – Electrical – Thermal/Magnetic – Mechanical – Soft)

Unit – III

- **Main**
  - IoT Networking
  - IoT Protocols
  - Wireless Technologies
- **Detail**
  - IoT networking basics
  - IoT components – Functional components
  - IoT service oriented architecture
  - IoT challenges

    - 6LowPAN – IEEE 802.15.4
    - ZigBee – RFID – NFC – Bluetooth
    - Wireless Sensor Networks – Applications

Unit – IV

- **Main**
    - Messaging Protocols
    - Application Protocols
    - Communication Models
- **Detail**
    - MQTT – Methods – Components
    - MQTT communication – Topics – Applications
    - SMQTT
    - CoAP – Message types – Request/Response model
    - XMPP
    - AMQP features – Components – Frame types

Unit – V

- **Main**
    - IoT Platforms
    - Cloud & Analytics
    - IoT Security
- **Detail**
    - Arduino – Raspberry Pi – Other IoT platforms
    - Data analytics for IoT
    - Cloud storage models – APIs
    - Attacks in IoT – Vulnerability analysis
    - IoT case studies (Smart Home – Smart farming)

CS802 (A) – Block Chain Technologies

Unit – I

- **Main**
  - Block Chain Fundamentals
  - Crypto Primitives
  - Security Aspects
- **Detail**
  - Block chain overview – Public ledger – Bitcoin – Smart contracts
  - Transactions – Distributed consensus
  - Public vs Private – Permissioned block chain
  - Hash function – Hash pointer – Merkle tree
  - Digital signature – Public key cryptography

Unit – II

- **Main**
  - Bitcoin Network
  - Consensus
  - Mining
- **Detail**
  - Bitcoin creation – Payments – Double spending
  - Bitcoin scripts – P2P network
  - Block mining – Block propagation
  - Proof of Work – Hash cash – Bitcoin PoW
  - Proof of Stake – Proof of Burn – Proof of Elapsed Time
  - Mining difficulty – Mining pool

Unit – III

- **Main**
  - Permissioned Block Chain
  - Consensus Models
  - Fault Tolerance
- **Detail**
  - Permissioned model – Use cases
  - Execute contracts – State machine replication
  - Paxos – RAFT
  - Byzantine problem – BFT systems

Unit – IV

- **Main**
  - Enterprise Applications

    - Finance & Trade
    - Identity
- **Detail**
    - Cross border payments – KYC
    - Food security – Mortgage
    - Trade finance – Supply chain
    - Identity on block chain

Unit – V

- **Main**
    - Block Chain Development
    - Smart Contracts
    - Platforms
- **Detail**
    - Hyperledger Fabric architecture
    - Identities – Access control – Channels
    - Smart contracts (Hyperledger – Ethereum)
    - Ripple – Corda

CS802 (B) – Cloud Computing

Unit – I

- **Main**
  - SOA & Web Services
  - Cloud Services
  - Virtualization
- **Detail**
  - SOAP – WSDL – UDDI – REST
  - SaaS – PaaS – Cloud benefits & limitations
  - Hypervisor – Monitoring cloud services

Unit – II

- **Main**
  - Utility Computing
  - Virtualization
  - Multi-tenancy
- **Detail**
  - Elastic computing – AJAX – Mashups
  - Virtualization in enterprises
  - Multitenant software – Cloud data stores

Unit – III

- **Main**
  - Cloud Data
  - MapReduce
  - Big Data Systems
- **Detail**
  - GFS – HDFS – BigTable – HBase – Dynamo
  - MapReduce model – Parallel processing
  - Applications of MapReduce

Unit – IV

- **Main**
  - Cloud Security
  - Privacy
  - Virtual Machine Security
- **Detail**
  - Cloud security architecture
  - Trusted cloud computing
  - VM threats – Security recommendations

Unit – V

- **Main**
  - Cloud Issues
  - Middleware
  - Resource Management
- **Detail**
  - QoS – Dependability – Data migration
  - Mobile cloud – Inter cloud
  - Load balancing – Monitoring – Performance evaluation

CS802 (C) – High Performance Computing

Unit – I

- **Main**
  - Processor Architecture
  - Memory Hierarchy
  - Optimizations
- **Detail**
  - Cache based architecture – Benchmarks
  - Pipelining – SIMD – Multicore
  - Vector processors – Compiler optimizations

Unit – II

- **Main**
  - Data Optimization
  - Parallel Computers
  - Network Systems
- **Detail**
  - Data access optimization
  - Shared – Distributed – Hybrid systems

Unit – III

- **Main**
  - Parallel Computing
  - OpenMP
  - Scalability
- **Detail**
  - Data – Functional parallelism
  - OpenMP execution – Synchronization
  - Load imbalance

Unit – IV

- **Main**
  - OpenMP Performance
  - Profiling
  - Optimization
- **Detail**
  - Performance pitfalls
  - Loop scheduling – False sharing

Unit – V

- **Main**
  - MPI Programming
  - Communication
  - Performance Tools
- **Detail**
  - Point-to-point – Collective communication
  - Virtual topologies
  - Communication overhead reduction

CS802 (D) – Object Oriented Software Engineering

Unit – I

- **Main**
  - OO Concepts
  - Paradigm
  - SDLC
- **Detail**
  - OO principles – Model architectures

Unit – II

- **Main**
  - RUP
  - Life Cycle
  - Workflows
- **Detail**
  - Best practices – 4+1 view model

Unit – III

- **Main**
  - UML
  - OO Analysis
  - Modeling
- **Detail**
  - UML diagrams – Tools
  - Use case – Activity – Class model

Unit – IV

- **Main**
  - OO Design
  - Behavioral Modeling
  - Case Studies
- **Detail**
  - CRC cards – Interaction – State charts
  - ATM – Payroll – Course systems

Unit – V

- **Main**

    - OO Testing
    - Project Management
    - Design Patterns
- **Detail**
    - Testing strategies
    - Rational tools
    - Introduction to design patterns

CS803 (A) – Image Processing & Computer Vision

Unit – I

- **Main**
  - CVIP Basics
  - Morphological Processing
  - Recognition
- **Detail**
  - Image filtering – Representation – Statistics
  - Dilation – Erosion – Opening – Closing
  - Region growing – Shrinking

Unit – II

- **Main**
  - Image Representation
  - Segmentation
  - Area Extraction
- **Detail**
  - Boundary – Region descriptors
  - Thresholding – Clustering – Split & merge
  - Hough transform – Curve fitting

Unit – III

- **Main**
  - Region Analysis
  - Boundary Analysis
  - Matching
- **Detail**
  - Spatial moments
  - Shape numbers
  - Structural matching – Model database

Unit – IV

- **Main**
  - Model Recognition
  - Projection
  - Object Matching
- **Detail**
  - Perspective geometry – Image matching

    - 2D models – Global vs Local features

Unit – V

- **Main**
    - Knowledge Based Vision
    - Feature Extraction
    - Machine Learning
- **Detail**
    - Object recognition – PCA
    - Neural networks – ML methods

CS803 (B) – Game Theory

Unit – I

- **Main**
  - Game Design
  - Meaningful Play
  - Design Process
- **Detail**
  - Game schemas – Iterative design
  - Discernable – Integrated play

Unit – II

- **Main**
  - Semiotics
  - Design Meaning
  - Interpretation
- **Detail**
  - Semiotic concepts – Context

Unit – III

- **Main**
  - Systems
  - Interactivity
  - Choice
- **Detail**
  - Open – Closed systems
  - Choice molecules

Unit – IV

- **Main**
  - Digital Games
  - Rules
  - Culture
- **Detail**
  - Magic circle – Game identity

Unit – V

- **Main**
  - Rules Design
  - Digital Rules

    - Case Studies
- **Detail**
    - Tic Tac Toe – Deck of Cards

CS803 (C) – Internet of Things (OE)

(same structure as CS801, converted already)

CS803 (D) – Managing Innovation & Entrepreneurship

Unit – I

- **Main**
  - Innovation
  - Entrepreneurship
  - Business Development
- **Detail**
  - Types of innovation
  - Competitive advantage

Unit – II

- **Main**
  - Sources of Innovation
  - Creativity
  - Management Methods
- **Detail**
  - Push – Pull – Analogies
  - Agile – Six Thinking Hats

Unit – III

- **Main**
  - Innovation Process
  - Open Innovation
  - Business Models
- **Detail**
  - Stage Gate method
  - In-house development

Unit – IV

- **Main**
  - Human Centered Innovation
  - Strategy
  - Co-creation
- **Detail**
  - Innovation strategies – Selection

Unit – V

- **Main**
  - Innovation Measurement
  - Failure Analysis
  - Workshops
- **Detail**
  - Financial – Non-financial metrics
  - Post-audit – Innovation workshop